# JYOTHIR RAGHAVALU BHOGI

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## EDUCATION

**Lovely Professional University** — Jalandhar, India
Bachelor of Technology (Computer Science and Engineering) — **July 2022 - May 2026**

## SKILLS SUMMARY

- **Programming**: Python, SQL
- **Machine Learning**: Machine Learning, Deep Learning, Speech Recognition Fundamentals, Audio Processing, Generative AI
- **Frameworks**: PyTorch, TensorFlow
- **Data**: Data Processing, Feature Engineering, Data Validation, Dataset Preparation
- **Model Engineering**: Model Training, Model Fine-Tuning, Model Optimization, Model Inference
- **Deployment & MLOps**: On-Device ML, Edge AI, Docker, CI/CD, Inference Optimization
- **Software Engineering**: Object-Oriented Programming (OOP), Data Structures, Debugging, Scalable Systems, SDLC, Agile
- **Tools**: Git, Docker
- **Collaboration**: Collaboration, Analytical Thinking, Problem Solving, Documentation, Communication

## WORK EXPERIENCE

**DATA ENGINEER INTERN | flickDONE | LINK** — **June 25- August 25**

- Optimized Python data pipelines processing 50,000+ daily records improving validation accuracy for ML pipelines.
- Engineered scalable SQL schema and indexing supporting efficient feature engineering for machine learning workflows.
- Developed automated validation scripts detecting anomalies early improving dataset quality for model training.
- Collaborated with ML engineers deploying production ML inference pipelines for speech/audio models.
- Implemented Docker CI/CD pipelines deploying on-device ML inference optimized for edge devices.

## PROJECTS

**Generative AI Product Recommender System | LINK** — **June 25- August 2025**

- Developed scalable recommender system using Python PyTorch SQL APIs enabling personalized product recommendations.
- Integrated generative AI and Large Language Models improving contextual recommendations and automation workflows.
- Implemented Docker CI/CD pipelines deploying machine learning inference services with optimized latency scalability.
- Designed API validation pipelines ensuring reliable predictions and stable deployment of recommendation models.
- Fine-tuned Whisper model for small dataset, improving real-time speech-to-text accuracy.
- Familiar with edge ML concepts and inference optimization for consumer hardware.

**Rails TODO App | LINK** — **September 24- November 2024**

- Developed full-stack task management platform using Ruby on Rails MVC architecture supporting backend development.
- Implemented REST APIs authentication systems and SQL schema improving security maintainability backend performance.
- Optimized debugging workflows and automated unit testing pipelines improving software reliability and efficiency.
- Ensured maintainable architecture through modular code design, documentation, and structured Agile development practices.

**Morning Market Brief | LINK** — **June 24- August 2024**

- Developed cloud analytics platform generating automated financial insights using scalable Python data pipelines.
- Built structured ETL workflows using Python and SQL enabling efficient dataset ingestion, transformation, and validation.
- Implemented REST API integration testing ensuring reliable data pipelines supporting machine learning workflows.
- Integrated Docker-based CI/CD pipelines enabling scalable deployment and improved system reliability across services.

## RESEARCH AND PUBLICATIONS

**Multi-Stage Ensemble for Driver Fatigue Recognition | IEEE, 2025**

- Developed PyTorch ensemble model for audio-based driver fatigue recognition using speech/audio signals.
- Optimized preprocessing pipelines using OpenCV techniques reducing model training computation and inference time.
- Applied hyperparameter tuning and reproducible validation workflows improving deep learning model accuracy significantly.

## CERTIFICATES

**Python, Data Science & ML (Cipherschools) | CERTIFICATE** — **August 2024**

- Mastered Python including object-oriented design, debugging, and scalable software engineering best practices.
- Applied machine learning algorithms feature engineering techniques and model evaluation strategies on datasets.

**Generative AI with LLMs (Coursera) | CERTIFICATE** — **March 2024**

- Learned transformer architectures embeddings and prompt engineering techniques for developing generative AI systems.
- Built practical Large Language Model applications enabling automated summarization text generation intelligent workflows.